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Find A Gene Project

The Find-A-Gene project must be completed in stages as a team:

1. Select a known, characterized protein of interest. This will be the QUERY PROTEIN. You are encouraged to select your favorite protein, one that is interesting/exciting to you or has relevance to your desired research area.

ALDH2_human is the gene we want to study.

Fasta Sequence: <https://www.ncbi.nlm.nih.gov/datasets/gene/217/>

Protein Accension: [NP_000681.2](#)

Tblastn:

https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=tblastn&PAGE_TYPE=BlastSearch&BLAST_SPEC=&LINK_LOC=blasttab&LAST_PAGE=blastp

2. Perform database searches to identify genomic DNA that encodes a new, NOVEL PROTEIN that is in the same family as your QUERY PROTEIN. Follow the steps below:
 - A. Perform a TBLASTN search with the QUERY PROTEIN sequence against a DNA database. This will convert your QUERY PROTEIN sequence from Stage 1 into six query DNA sequences based on the six different reading frames and search for all six queries in the chosen DNA database. To find a new gene, search databases likely to contain uncharacterized sequences, such as experimental data repositories and genomic DNA databases. **Your best bet is the WGS database.** The WGS database contains data from organisms currently being sequenced and has a significant amount of non-annotated sequence data. You do have to select some type of organism to do a WGS search, but you can pick a taxonomic classification that includes multiple species, e.g. bony fishes, primates, etc. The search will go very slow unless you limit the number of species that you are searching. Thus, this step involves a bit of trial and error as you pick and choose different taxonomic groups to explore in search of the NOVEL PROTEIN that is related to the QUERY PROTEIN. The "Taxonomic Groups" menu on the left-hand side of the WGS Browser

- B. Links to an external site. allows you to explore different taxa and see how many WGS projects are associated with a particular group, enabling you to select a search space of limited size. Also, selecting taxonomic groups with less "popular" organisms (avoid humans, mice, etc.) will give you a better chance of finding a never before annotated gene.

Try selecting *Microcebus murinus* (Gray mouse lemur - taxid: 30608) in the WGS database as the target organism.

[WP5V9GF1014](#)

***Microcebus ravelobensis* (taxid:122231) - Golden Brown Mouse Lemur**

<https://blast.ncbi.nlm.nih.gov/Blast.cgi?CMD=Get&RID=WS3ZB75W0>

QID: [WS4PR2XD016](#)

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- C. Inspect the output of the TBLASTN BLAST search from Part A. There are three types of hits you may find.
- I. False positive DNA sequences that are not homologous to the query protein sequence. These hits generally have high E values and weak or no statistical significance. These are not NOVEL PROTEIN candidates.
 - II. DNA sequences encoding known genes that are not novel. These hits are labeled or annotated with the name of your query protein/gene. Annotated hits are not new. You should be especially wary if the hits are in the same species as your query. Also, known genes often are identical or near-identical to your protein query with very low E-values. These are not NOVEL PROTEIN candidates.
 - III. DNA sequences encoding novel genes. These hits are not annotated, often are labeled with the genome/chromosome to which they have been mapped, and often associated with a species different than the species from which your query is derived. Generally, novel gene hits have low and statistically significant E values. These are NOVEL PROTEIN candidates.

Hypothetical proteins count as NOVEL PROTEIN candidates if the hypothetical protein is not listed as an ortholog on the NCBI Gene database entry for the QUERY PROTEIN. GenBank uses the classification hypothetical for proteins that have no evidence supporting them except for DNA alignments.

- D. NOVEL PROTEIN candidates must align to at least 80% of the QUERY PROTEIN. The "Query Cover" statistic must be >80%. I want you to analyze full protein sequences, not partial sequences.

Candidate 1 / Chosen Candidate

[VSMG01017660.1](#)

- E. Once you identify some candidate NOVEL PROTEIN candidates, you need to obtain the DNA and protein sequences of the candidates. Depending on the hit type, there are a few ways to do this from the BLAST results page.
- I. If your NOVEL PROTEIN candidate was listed as a hypothetical protein, then click the link in the "Accession" column of the "Descriptions" tab of the BLAST results. This will take you to the GenBank file page describing the genomic sequence you found in the WGS (these pages can contain a lot of sequencing data and may take a few seconds to load). You can search this page for the name of your QUERY PROTEIN, which may show as "similar to" a hypothetical protein. If you find a matching hypothetical protein, then the NOVEL PROTEIN sequence will be shown in the "translation" section of the "CDS" feature. The DNA/genome sequence can be generated by clicking the "CDS" feature and displaying it as a "FASTA" sequence.
 - II. If your NOVEL PROTEIN candidate was listed as a hypothetical protein but is not labeled as "similar to" your QUERY PROTEIN, then click the link in the "Description" column on the "Descriptions" tab. If you performed the BLAST search against the WGS database, then you were searching genomic DNA. Genes are broken up into exons and introns in genomic DNA. Thus, your QUERY PROTEIN likely aligned to multiple regions of the genomic DNA

sequence, as proteins are made up of only exon DNA. This will show up as different alignments labeled "Ranges" for your candidate novel gene. Next to one of the Range labels, click the "GenBank" link. This may take you a GenBank file page corresponding to a hypothetical protein in the genomic DNA. The full NOVEL PROTEIN sequence is found at the bottom of the page. You can then take part of this NOVEL PROTEIN sequence and search for it in the GenBank file page found by clicking the link in the "Accession" column of the "Descriptions" tab, which contains the entire genomic DNA sequence hit. You should be able to find the NOVEL PROTEIN sequence on this page, still labeled as hypothetical but not labeled as similar to your QUERY PROTEIN. You can generate the DNA/genome sequence by clicking the "CDS" feature and displaying it as a "FASTA" sequence.

1. If your NOVEL PROTEIN candidate is not listed as a hypothetical protein, then the NOVEL PROTEIN will not show up as a feature on the GenBank file page linked to the "Accession" column of the "Descriptions" tab of the BLAST results. This means you are stuck downloading only the sequences that aligned to your QUERY PROTEIN on the BLAST results page. Click the link in the "Description" column on the "Descriptions" tab. Then, click "Download" > "FASTA (aligned sequences)." This will download all the DNA sequences that code for the NOVEL PROTEIN candidate that aligned to your QUERY PROTEIN. Again, there are likely multiple DNA sequences for a single hit, because the exons coding for protein are split up by introns, which do not code for amino acids and will not align to the QUERY PROTEIN. Assemble as much of the gene as possible from the partial sequences. The Range coordinates on the BLAST results (also shown in the label of the downloaded FASTA sequences) will guide you into which order to assemble the gene fragments. Ignore fragments that are way out of range or that overlap larger fragments. The

assembled sequence represents the gene encoding your NOVEL PROTEIN. Translate this DNA sequence into a protein sequence using any online tool.

Golden Brown Mouse Lemur Protein Sequence:

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FQLGSPWRRMDASSRGRLNLNRLADLIERDRTYLAALETLDNGKPYVISYLVLDLDMVLKCLRYYAGWADK
YHGKTIPIIDGFFSYTRHEPVGVCQIIPWNFPLLMQAWKLGPALATGNVVVMKVAEQTPLTALYVANL
IKEAGFPPGVVNIVPGFGPTAGAAIASHEDVDKVAFTGSTEIGRLIQVAAGSSNLKRVTTLELGGKSPNI
IMSDADVDWAVEQAHFALFFNQGCCAGSRTFVQEDVYNEFVERSVARAKSRVVGPNPFDKTEQGPQV
DETQFKKILGYINSGKQEGAKLLCGGGPAADRGYYIQPTVFGDVQDGMTIAKEEIFGPVMQILKFKTIE
EVVGRANNSNYGLAAVFTKDLDKANYLSQALQAGTVWVNCYDVFGAQSPFGGYKMSGSGRELGEYGLQ
AYTEVKTVST 510

Master DNA Sequence Excluding Gaps

ATGTTTCAAT	TAGGTTCTCC	TTGGCGTCGT	ATGGATGCTT	CTTCTCGTGG	TCGTTTATTA
AATCGTTTAG	CTGATTTAAT	TGAACGTGAT	CGTACTTATT	TAGCTGCTTT	AGAAACTTTA
GATAATGGTA	AACCTTATGT	TATTTCTTAT	TTAGTTGATT	TAGATATGGT	TTTAAAATGT
TTACGTTATT	ATGCTGGTTG	GGCTGATAAA	TATCATGGTA	AAACTATTCC	TATTGATGGT
GATTTTTTTT	CTTATACTCG	TCATGAACCT	GTTGGTGTTT	GTGGTCAAAT	TATTCCTTGG
AATTTTCCTT	TATTAATGCA	AGCTTGGAAG	TTAGGTCCTG	CTTTAGCTAC	TGGTAATGTT
GTTGTTATGA	AAGTTGCTGA	ACAAACTCCT	TTAACTGCTT	TATATGTTGC	TAATTTAATT
AAAGAAGCTG	GTTTTTCCTC	TGGTGTGTGT	AATATTGTTC	CTGGTTTTGG	TCCTACTGCT
GGTGCTGCTA	TTGCTTCTCA	TGAAGATGTT	GATAAAGTTG	CTTTTACTGG	TTCTACTGAA
ATTGGTCGTT	TAATTCAAGT	TGCTGCTGGT	TCTTCTAATT	TAAAACGTGT	TACTTTAGAA
TTAGGTGGTA	AATCTCCTAA	TATTATTATG	TCTGATGCTG	ATGTTGATTG	GGCTGTTGAA
CAAGCTCATT	TTGCTTTATT	TTTAAATCAA	GGTCAATGTT	GTTGTGCTGG	TTCTCGTACT
TTTGTTCAAG	AAGATGTTTA	TAATGAATTT	GTTGAACGTT	CTGTTGCTCG	TGCTAAATCT
CGTGTTGTTG	GTAATCCTTT	TGATTCTAAA	ACTGAACAAG	GTCCTCAAGT	TGATGAAACT
CAATTTAAAA	AAATTTTAGG	TTATATTAAT	TCTGGTAAAC	AAGAAGGTGC	TAAATTATTA
TGTGGTGGTG	GTCCTGCTGC	TGATCGTGGT	TATTATATTC	AACCTACTGT	TTTTGGTGAT
GTTCAAGATG	GTATGACTAT	TGCTAAAGAA	GAAATTTTTG	GTCCTGTTAT	GCAAATTTTA
AAATTTAAAA	CTATTGAAGA	AGTTGTTGGT	CGTGCTAATA	ATTCTAATTA	TGGTTTAGCT
GCTGCTGTTT	TTACTAAAGA	TTTAGATAAA	GCTAATTATT	TATCTCAAGC	TTTACAAGCT
GGTACTGTTT	GGGTTAATTG	TTATGATGTT	TTTGGTGCTC	AATCTCCTTT	TGGTGGTTAT
AAAATGTCTG	GTTCTGGTCG	TGAATTAGGT	GAATATGGTT	TACAAGCTTA	TACTGAAGTT
AAACTGTTT	CTACTTAA				

III. Check the results of the translation. If there is a stop codon in your translated protein sequence, then you might have an intron sequence present. You should go back and delete the fragments containing the intron or try to remove the intron. Introns almost always start with "GT" and end with "AG." Once you delete the appropriate fragment or intron from your NOVEL PROTEIN gene sequence, re-translate the exon only sequence into a protein sequence.

F. To confirm that your NOVEL PROTEIN candidate is truly novel, you must use the NOVEL PROTEIN sequence as a BLASTX query against the NCBI Non-redundant protein sequences (nr) database. The BLASTX search will convert your novel gene/DNA sequence into six query protein sequences based on the six different reading frames and search for all six queries in the premier protein database on the NCBI website. If there is a match with >99% amino acid identity to a protein in the database from the same species as your NOVEL PROTEIN candidate, then your NOVEL PROTEIN candidate is NOT novel unless the hit is labeled as hypothetical and is not listed as an ortholog on the NCBI Gene database entry for the QUERY PROTEIN. **Ideally, the BLASTX results would show no hits in the target organism but related hits in other organisms. This is strong evidence that you found a new gene for your target organism.** But you may also find hits in the same species with significantly less than 99% identity to your query. These require further investigation to determine their novelty; you need to make sure your novel gene candidate is not simply a sequence variant of a characterized gene. If there are no database matches from the BLASTX search, you have found a gene/protein that is not homologous to the QUERY PROTEIN and should try a different candidate from initial search or go back and repeat the initial TBLASTN search or change the QUERY PROTEIN.

I. For many initial TBLASTN searches with a QUERY PROTEIN of interest, it is easy to find "novel" genes; for some cases it is not easy to find new genes, perhaps because relevant homologs do not exist or because the appropriate database is not searched.

3. Gather more information about the protein family (the same family should include your QUERY and NOVEL PROTEINS). Interesting questions include: what are the functions of the proteins? How many family members have been identified? Are family members found in all organisms or just certain taxa? What does the 3D structure look like for proteins in this family? There are a few strategies to find this information in protein family databases (detailed below), but be careful here. Some protein family databases focus on certain protein domains shared among non-related proteins and not on full-length, evolutionarily-related proteins. Be precise in describing the information you find.

Function:

“Required for clearance of cellular formaldehyde, a cytotoxic and carcinogenic metabolite that induces DNA damage.” - InterPro, (<https://www.uniprot.org/uniprotkb/P05091/entry>)

Total Number of Family Members:

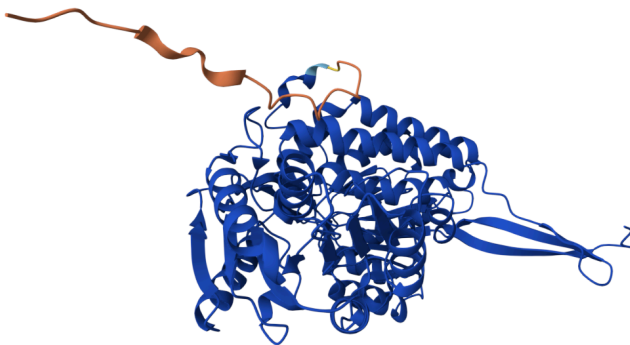
404,0000

Found in All Organisms or Certain Taxa:

All organisms

<https://www.ebi.ac.uk/interpro/entry/pfam/PF00171/taxonomy/uniprot/#sunburst>

3d Structure Example:



(<https://alphafold.ebi.ac.uk/entry/AF-P05091-F1>)

- A. Search the protein family databases with the name of your query protein or find links to protein family databases from the Gene or UniProt pages describing your QUERY PROTEIN.

<https://www.uniprot.org/uniprotkb/P05091/entry>

- B. If you found a hypothetical protein, then the GenBank pages describing the hypothetical protein may link directly to protein family databases, like the CDD, InterPro, and Pfam.

No we did not

- C. Search InterPro, the premier protein family database, using your QUERY PROTEIN.

<https://www.ebi.ac.uk/interpro/result/InterProScan/iprscan5-R20260331-194356-0625-57141072-p2m/internal-1774982630916-118-1/>

Interpro Novel Sequence Search: [iprscan5-R20260331-194724-0735-42260155-p1m](https://www.ebi.ac.uk/interpro/result/InterProScan/iprscan5-R20260331-194724-0735-42260155-p1m)

- D. You may also perform a powerful InterPro search using the NOVEL PROTEIN sequence. Use the "Search by sequence" function to search InterPro using only the sequence of your NOVEL PROTEIN. This is particularly helpful when trying to figure out which InterPro entry best describes the protein family or when you have a partial NOVEL PROTEIN sequence and want to know what domains are represented in the partial sequence.

Optional

<https://www.ebi.ac.uk/interpro/result/InterProScan/iprscan5-R20260331-200725-0841-74292952-p1m/>

<https://www.ebi.ac.uk/interpro/result/InterProScan/iprscan5-R20260331-194356-0625-57141072-p2m/>

4. Generate a multiple sequence alignment that includes your novel protein, your original query protein, and a other members of the protein family. The multiple alignment should include 10-30 protein sequences. Specify any parameters that you change from the default settings. **You should try at least two different multiple sequence alignment algorithms and compare their results. Pick one alignment to move forward to the next stage.**

MUSCLE, accurate and good for proteins:

<https://www.ebi.ac.uk/jdispatcher/msa/muscle/summary?jobId=muscle-I20260401-200127-0077-53357814-p1m>

Clustal OMEGA, scalable progressive alignment:

<https://www.ebi.ac.uk/jdispatcher/msa/clustalo/summary?jobId=clustalo-I20260401-200115-0789-70748613-p1m>

- A. When selecting orthologs for the alignment, pick well-known, traditional species, e.g. humans, mice, etc., but also pick some species that are similar to the species that has your novel gene. For example, if you find a novel gene in a violet, then add some plants/flowers to your alignment. Ortholog protein sequences can be found in many databases, including the NCBI Gene page, UniProt, and any of the protein family databases.
 - B. If your novel protein sequence is a partial sequence, then delete all the columns in the alignment that do not include your partial sequence when using the alignment as an input for the phylogenetics analysis. This can be done quite easily after converting the alignment into MEGA format. It is very important to do this, as the partial protein will be shown to have huge gaps at the ends of the sequence that will affect the output of the phylogenetics analysis.
5. Generate a phylogenetic tree showing the relation of the new protein to the other related proteins that were part of the multiple sequence alignment. To do a proper phylogenetics analysis, complete the following steps:
- A. Determine the best substitution model for your input multiple sequence alignment using the "Find Best DNA/Protein Models (ML)" function in MEGA.
 - B. Generate a phylogenetic tree using the best substitution model for your data as determined by the previous step. Run a bootstrapping analysis to provide statistical support for your tree. Tree rooting is optional. Specify any parameters that you change from the default settings. **You should try at least two different tree building algorithms and compare their results.**

